

ELSD-Net: A Novel Efficient and Lightweight Ship Detection Network for SAR Images

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Abstract—Deep learning has shown significant promise in synthetic aperture radar (SAR) ship detection due to its powerful target recognition capabilities. However, SAR ship detection still faces unique challenges, including low image resolution, difficulty in detecting small targets, and the dense clustering of inshore ships. Additionally, traditional models tend to be complex and computationally intensive, making deployment on satellite platforms difficult. To address these issues, nonstride feature extraction (NSFE) module is proposed to construct the backbone network, significantly reducing the number of parameters, and multiscale efficient channel attention (MECA) is designed to further enhance the multiscale perception of the module. Moreover, strip partial convolution (SPC) module is designed to improve global modeling capability of the network with fewer parameters. Taking full advantage of the above modules and based on YOLOv5n architecture, we propose a novel efficient and lightweight ship detector named ELSD-Net for SAR images. Experimental results on the SAR ship detection dataset (SSDD) demonstrate that ELSD-Net achieves high detection accuracy 98.2% with fewer parameters 0.3 M and lower Flops 1.2 G compared to other methods.

Index Terms—Lightweight, ship detection, synthetic aperture radar (SAR), you only look once (YOLO).

I. INTRODUCTION

SYNTHETIC aperture radar (SAR), as an active microwave radar, is less affected by weather conditions, which make it possible to work all day and all weather [1]. Therefore, SAR images have extremely important application value for ocean monitoring. With the proposal and full development of deep learning, the use of deep learning to achieve ship detection has attracted wide attention [2].

Up to now, many deep learning object detection methods have been proposed, which can be roughly divided into two categories: convolutional neural network (CNN) based and transformer based. Among them, CNN can be subdivided into anchor-based method and anchor-free method, and anchor-based method can be further subdivided into two-stage method and single-stage method. The former, like mask R-CNN, divides target detection task into candidate region selection and target regression. It can accomplish high

detection accuracy, but typically incur a considerable computational cost. The latter, like SSD, enables end-to-end detection, reducing model size at a slight expense of detection accuracy, but fortunately various researches are gradually narrowing this accuracy gap. Considering the accuracy of ship detection and the strict computational power distribution required for deployment on satellites, single-stage method is the most appropriate.

In the process of continuous optimization of the above methods, deep learning has made great progress in the field of SAR image ship detection [3], [4], [5], [6], [7]. Zhou et al. [8] proposed a lightweight metalearning-based SAR image few-shot object detection method with few labeled data. Tian et al. [9] used inception depthwise convolution to design a faster and lighter ship detector. Sun et al. [10] proposed bidirectional feature fusion and angular classification based on YOLO to improve detection performance of arbitrary-oriented ships. Xiong et al. [11] redesigned the pyramid pooling structure for ship detection and recognition in complex-scene SAR images. Moreover, some researches focus on utilizing the special characteristics of SAR images for target detection. Joshi et al. [12] designed a real-time ship detection strategy for range-compressed radar data. Leng et al. [13] leveraged the differences in how ships and clutter respond to radar signals, and proposed a novel ship detection method for raw SAR echo data.

Although these research works have made abundant progress, there are still opportunities for further improvement. First, reducing the training parameters of the model is a significance research topic due to the cost associated with deploying models on satellite platforms. Meanwhile, compared with optical images, clarity and detail information of SAR images are often unsatisfactory, so the influence of background clutter noise needs to be suppressed. In addition, the small size of the ship and the dense arrangement near the shore make the multiscale target and complex scene detection of the ship in SAR images particularly important.

To address these challenges, we design an efficient and lightweight ship detection network named ELSD-Net based on YOLOv5n, which is inspired by space-to-depth convolution (SPDConv) [14], MobileNetv3 [15], efficient channel attention (ECA) [16], and partial convolution (PConv) [17]. The designed model reduces the number of parameters, improves the detection performance of multiscale ship targets, mitigates the background noise interference, and enhances the global sensing capability.

Received 4 November 2024; revised 30 December 2024; accepted 6 January 2025. Date of publication 14 January 2025; date of current version 30 January 2025. (Corresponding author: Weidong Yu.)

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Digital Object Identifier 10.1109/LGRS.2025.3527444

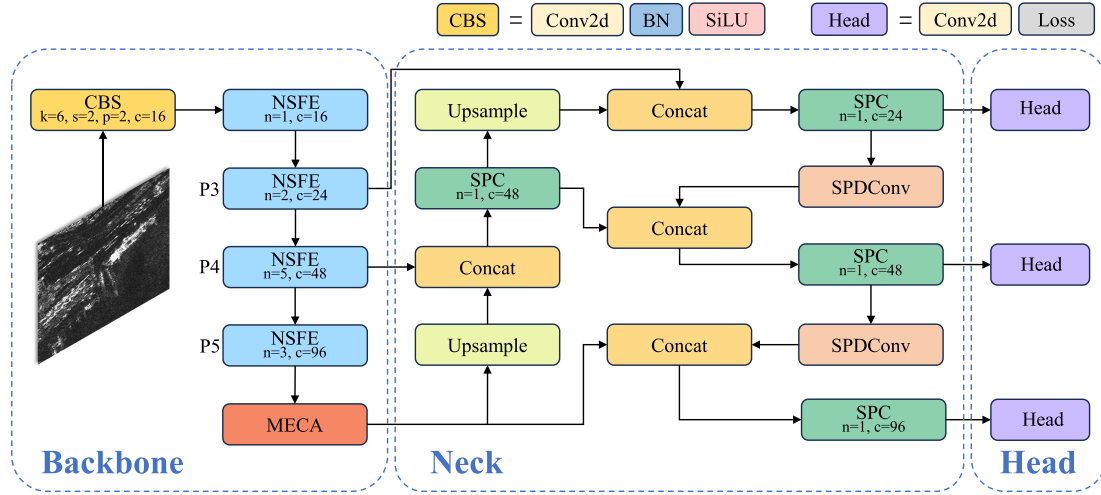


Fig. 1. Overall architecture of ELSD-Net.

The primary contributions of this article are shown as follows.

- 1) In order to achieve considerable lightweight of the model, we propose NSFE module based on SPDConv and MobileNetv3 to enhance the feature extraction capability of the model, ensuring the detection accuracy while drastically reducing the number of parameters and flops.
- 2) Based on the common ECA module, we designed the MECA module, which integrates global and local information in the attention mechanism, enhances the model's multiscale perception of features, and further improves the detection accuracy without increasing the complexity of the model.
- 3) The designed SPC module, which uses two orthogonal strip convolutions as an approximation to the standard large kernel convolution, is lightweight to reduce interference from neighboring targets by capturing remote contextual information.

II. NETWORK DETAILS

In this section, the overall structure of network, the training, and inference process are first presented. Information on the specific details of each designed module will then be presented separately.

A. ELSD-Net Structure

The structure of ELSD-Net is shown in Fig. 1, which is improved on YOLOv5n. During training process, SAR images first undergo data enhancements, including mosaicking, blurring, and flipping. The enhanced images are then fed into backbone consisted of NSFE and MECA, generating three feature maps of different sizes P3, P4, P5. The generated feature maps further interact semantic information for feature fusion in neck consisted of SPC. Finally, the fused features are passed through three detection heads to produce outputs that predict large, medium, and small ship targets on the corresponding feature maps. During inference process, data enhancements are no longer be used. The detection results obtained from

heads are decoded and the model performances are evaluated. Further details of the modules used in ELSD-Net will be found below.

B. NSFE Module

The root cause of the difficulty in detecting small targets in CNN architectures can be the common strided convolution or pooling layer [14], which are widely used in downsampling, and the adverse effects of this design usually do not show up because many research scenarios are friendly, the images have good resolution, and the targets are of moderate size. However, in complex tasks where the images are blurry or the objects are very small, such as SAR images ship detection, there is not enough redundant information for learning, and the model begins to suffer from the loss of fine-grained information.

To address this problem, and considering the need for lightweight extraction of features, we combine the advantages of SPDConv and MobileNet to design the NSFE module, the specific structure of which is shown in Fig. 2. Considering any feature map F of size $H \times W \times C$, it is first sliced into four feature submaps of size $H/2 \times W/2 \times C$ according to the following equation:

$$\begin{aligned}
 f_{0,0} &= F[0:2:H, 0:2:W] \\
 f_{0,1} &= F[0:2:H, 1:2:W] \\
 f_{1,0} &= F[1:2:H, 0:2:W] \\
 f_{1,1} &= F[1:2:H, 1:2:W].
 \end{aligned} \tag{1}$$

Further, four feature submaps are spliced along the channel dimension, followed by a nonstride 1×1 convolution to obtain a new feature map F_{new} of size $H/2 \times W/2 \times C$, so that the downsampling is completed without applying strided convolution or pooling layer. All feature information is retained in the channel dimension, which improves the adverse effect of information loss.

The new feature map F_{new} then extends channels by the CBS module. Next, the depthwise separable convolution is applied to extract features, while using the channel attention mechanism. Finally, the number of channels is compressed, and a long-hop connection connects with F_{new} to produce

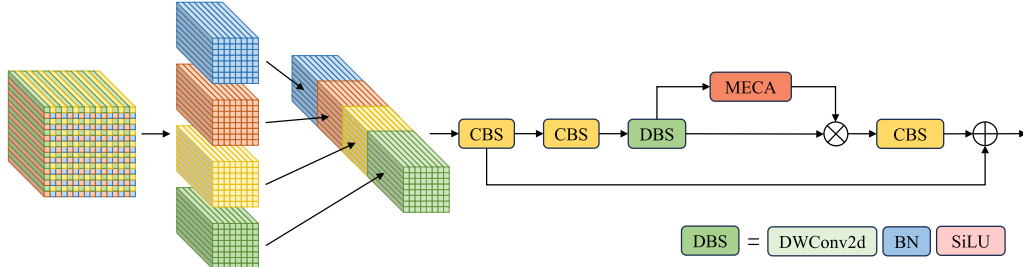


Fig. 2. Structures of the NSFE.

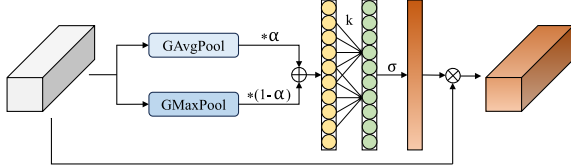


Fig. 3. Structures of the MECA.

the output, extracting more feature information while using fewer modules. Overall, the designed NSFE module mitigates the negative impact of strided convolutions on the detection of small ships for SAR images and significantly reduces the number of parameters required for feature extraction.

C. MECA Module

SAR images are limited by their imaging mechanism, and susceptible to background clutter noise. Although it is difficult to suppress the noise directly, we can still indirectly suppress the noise by focusing more attention on the ship through the appropriate attention mechanism. Based on the ECA, the MECA module is designed as shown in Fig. 3, which can effectively suppresses the background clutter noise with minimal operation overhead.

The MECA module first performs global average pooling and global maximum pooling on each channel of the input feature map I and multiplies them by the learnable parameters α and $(1 - \alpha)$, respectively, before summing them up, where α has an initial value of 0.5 and ranges in $(0, 1)$. Next, MECA generates channel weights W by performing a fast 1-D convolution of size k and passing through the sigmoid activation function. The final output is $O = I \times W$. MECA enhances the model's multiscale perception of features. This is because average pooling considers the information of the whole feature map and helps to capture the global perspective information, while maximum pooling highlights the salient features in the feature map and helps to extract the local detail information. Summing the results of these two pooling operations can fuse global and local information in the attention mechanism, thus improving the expressiveness and performance of the model. At the same time, the learnable parameters give the model the ability to adaptively and dynamically adjust whether more globally or locally focused.

D. SPC Module

There is another inherent defect of CNN architecture, that is the lack of global modeling capability, which is not conducive to the ship detection of SAR images in complex scenes such as nearshore or dense arrangement. Although this shortcoming

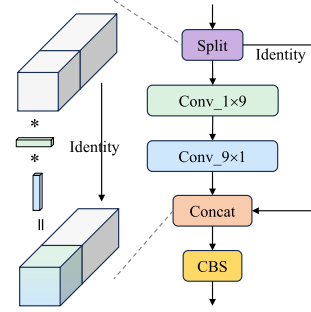


Fig. 4. Structures of the SPC.

can be mitigated by using large kernel convolution, the complexity of the network model increases accordingly. In order to optimize the global modeling capability of the network, SPC module shown in Fig. 4 is designed to capture remote contextual information.

The SPC module uses two sets of orthogonal strip convolutions as a simplified version of the standard large kernel convolution. This design first considers that for an approximate long strip-shaped target such as the ship, strip convolution can obtain the interdependence among pixels at a long distance, and more conveniently extract and fuse the features. Furthermore, strip convolution is lightweight, when compared with the traditional 2-D convolution kernel. Assuming the convolution kernel size is K , both parameter and computational complexity are reduced to $2/K$ of standard large kernel convolution. So, two orthogonal 1-D convolutions can be used to achieve a similar effect without significantly increasing the computational cost like the standard large kernel convolution. Moreover, inspired by PConv in design, SPC module applies the orthogonal strip convolution described above to only part of the input channels and keeps the remaining channels untouched, considering the redundancy in feature mapping. Both parts are then pointwise convolved through the CBS to ensure that the feature information flows among all channels. This design ultimately has the effect of reducing redundant computation and achieving better detection performance.

III. EXPERIMENTS

In this section, the experimental setup and data information will be given first. Then, the evaluation metrics will be presented. Finally, ablation experiments and comparison experiments will be performed on the experimental results.

A. Experimental Setup

The same computer configuration is used for all experiments. The GPU of the computer is RTX2050 (4G), the CPU

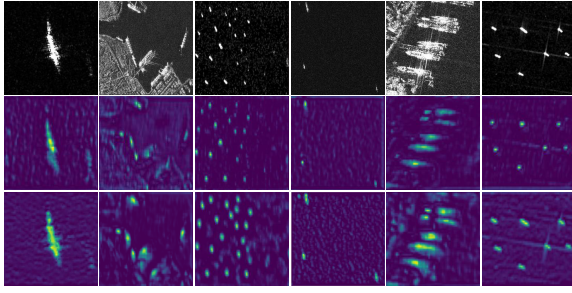


Fig. 5. Feature visualization. First row shows the original SAR images, second row shows the feature visualization of YOLOv5n, and last row shows the feature visualization of ELSD-Net.

is a 12th Gen Intel Core i5-12500H with 2.50 GHz, and the operating system used is Windows 11. The software versions in use include Python 3.9, CUDA 11.8, and PyTorch 2.0.

In order to ensure the fairness and flow of the experimental verification, publicly available SAR ship detection dataset (SSDD) [18] is used to verify ELSD-Net. The SSDD, which contains 1160 images of 2456 ships, covers both coastal and offshore scenarios, and the images are obtained from the RadarSat-2, Terra, and SAR-XSentinel-1 satellites.

For training, we follow the general convention of taking 20% of the images (with suffixes 1 or 9) for testing and the rest images for training. Adam is used as the optimizer and 300 epochs are trained, with the input image size of 640×640 and the batch size of 16. The initial learning rate is 0.01 and the learning rate is dynamically adjusted during training process.

B. Evaluation Indicators

The mean average precision (mAP) when $\text{IoU} = 0.5$, AP_{small} (area $\leq 32 \times 32$) when $\text{IoU} = 0.5$: 0.05: 0.95, floating point operations (FLOPs), parameters, and inference time per image are used to verify the detection performance of ELSD-Net

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

$$\text{mAP} = \int_0^1 P(R) \cdot dR \quad (3)$$

where TP means the true positive, FP means the false positive, and FN means the false negative. Because ship detection is single-category target detection, mAP is equivalent to AP.

C. Ablation Experiments

Ablation experiments are conducted on SSDD to verify the effectiveness of NSFE, MECA, and SPC modules in improving detection performance, and YOLOv5n is selected as the baseline for comparison. As shown in Table I, the experimental results of each step are compared to the baseline, and different colors are used to mark the change in performance, with red indicating the improvement over the baseline and blue indicating the decrease.

When NSFE module is used to rebuild the backbone, the model complexity is greatly optimized at the cost of a slight decrease in detection accuracy. Parameters and flops decrease by about 70%, which demonstrates that the NSFE can effectively extract features while significantly reduce the

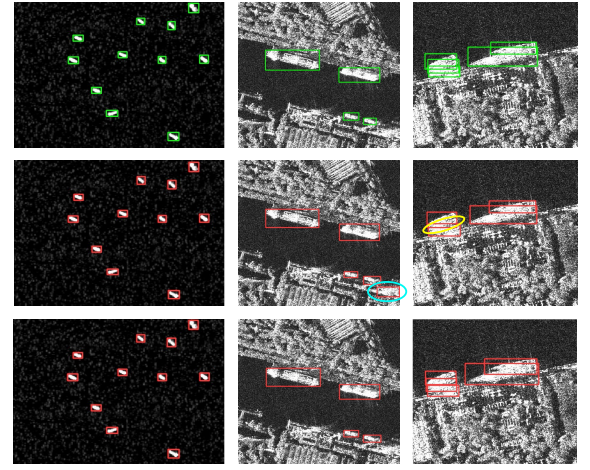


Fig. 6. Detection results visualization. First row shows the ground truth, second row shows the detection results visualization of YOLOv5n, and last row shows the detection results visualization of ELSD-Net. Green: ground truth. Red: detection results. Blue: false alarms. Yellow: missed ships.

complexity of the network. The addition of MECA module to the backbone improves detection accuracy with almost no increase in parameters and this improvement is also effective for small ship targets, which validates the feasibility of the method that uses the attention to indirectly suppress noise and make the model focus more on ship features. After the neck network is reconstructed with SPC module, the detection performance is further improved. Collectively, compared to YOLOv5n, mAP is improved by 0.8% and AP_{small} is improved by 1.4%. This performance improvement does not require additional computational cost. In fact, parameters decrease by about 82% and flops decrease by about 71%, which once again proves the effectiveness of the proposed method.

Fig. 5 shows the feature visualization of YOLOv5n and ELSD-Net in different SSDD scenarios at different scales, demonstrating that the proposed module effectively enhances the network's attention on ship targets. The visible comparison of YOLOv5n and ELSD-Net for ship detection is shown in Fig. 6. It can be seen that ELSD-Net has excellent ship detection ability and has lower false alarms and missed detections in complex environments such as inshore scene and the dense clustering of ships.

D. Comparison Experiments

To further demonstrate the excellent performance of ELSD-Net, we selected representative models from mainstream target detectors for performance comparison, including two-stage Mask RCNN, single-stage SSD, anchor-free FCOS, and baseline YOLOv5n. In addition, to ensure comprehensiveness of the comparison, we also compare improved methods designed for SAR images ship detection, including LPEDet [19], LMSD-YOLO [20], MSSD-Net [21], and MHASD [22].

The results of the comparison are shown in Table II. It is clear that ELSD-Net outperforms mainstream target detectors in all aspects on the SSDD dataset. Moreover, compared with the improved methods, the proposed method achieves the highest detection accuracy, the fewest parameters and the fewest flops, although the inference time is slightly insufficient. Therefore, the designed ELSD-Net is superior to the current

TABLE I
RESULT OF ABLATION EXPERIMENTS ON SSDD DATASET

NSFE	MECA	SPC	mAP(%)	AP _{small} (%)	P(%)	R(%)	Parameters(M)	FLOPs(G)
×	×	×	97.4	65.3	95.8	94.2	1.7	4.1
✓	×	×	97.1 (-0.3)	65.2 (-0.1)	95.6 (-0.2)	93.5 (-0.7)	0.4 (-76.5%)	1.3 (-68.3%)
✓	✓	×	98.0 (+0.6)	66.2 (+0.9)	96.1 (+0.3)	93.9 (-0.3)	0.4 (-76.5%)	1.3 (-68.3%)
✓	✓	✓	98.2 (+0.8)	66.7 (+1.4)	96.5 (+0.7)	94.1 (-0.1)	0.3 (-82.4%)	1.2 (-70.7%)

TABLE II
COMPARISON OF DIFFERENT SHIP DETECTION METHODS ON
THE SSDD DATASETS

Method	mAP(%)	Parameters(M)	FLOPs(G)	Time(ms)
Mask-RCNN	90.7	91.6	214.3	59.4
SSD	91.2	23.8	43.9	46.5
FCOS	96.8	32.3	72.4	22.7
YOLOv5n	97.4	1.7	4.1	10.5
LPEDet*	97.4	5.7	18.4	7.0
LMSD-YOLO*	98.0	3.5	6.6	14.7
MSSD-Net*	98.0	1.6	4.8	-
MHASD*	97.4	5.5	13.7	5.3
ELSD-Net	98.2	0.3	1.2	9.8

Note: The experimental data of methods with * are from their original articles.

detector in accuracy and complexity, and can be competent for the task of SAR images ship detection with satisfactory results.

IV. CONCLUSION

This research proposes the ELSD-Net, a ship detector for SAR images, which achieves lightweight ship detection effectively. The proposed method fully considers the uniqueness of SAR ship images, and makes improvements to address the inherent problems such as small ship target, low SAR image clarity and dense ship arrangement. NSFE module is designed to build the lightweight backbone to extract features with extremely low parameters and flops. MECA module is proposed to indirectly optimize the effect of background clutter noise. SPD module is designed to reconstruct neck to obtain remote context information and enhance global sensing capability of the network. The ablation experiments verify the effectiveness of all proposed modules, and the comparison experiments moreover demonstrate that the ELSD-Net achieves better performance and can complete the ship detection task satisfactorily. Moreover, the results can be extended to subsequent tasks such as classifying ship categories, as the details of ship targets become more distinct in high-resolution SAR images. In the future, we plan to further refine the network to handle more complex ship detection and classification tasks.

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